

STA 5635 HW 5

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February 6, 2026

Problem 1

- (1a) Loss function: $L(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N f(y_i \mathbf{x}_i^T \mathbf{w}) + s \|\mathbf{w}\|_2^2$, where $f(x) = \max(0, 1 - x)$. I do not believe that the derivative of $f(x)$ exists, at least not cleanly. The portion of the loss consisting of the average of the function only contributes if the function doesn't return 0. In other words:

$$f(y_i \mathbf{x}_i^T \mathbf{w}) = \begin{cases} 0 & y_i \mathbf{x}_i^T \mathbf{w} \geq 1 \\ 1 - y_i \mathbf{x}_i^T \mathbf{w} & y_i \mathbf{x}_i^T \mathbf{w} < 1 \end{cases}$$

So for a point where $y_i \mathbf{x}_i^T \mathbf{w} < 1$, the gradient is $\frac{\partial}{\partial \mathbf{w}}(1 - y_i \mathbf{x}_i^T \mathbf{w}) = -y_i \mathbf{x}_i^T$. For a points where $y_i \mathbf{x}_i^T \mathbf{w} \geq 1$, the gradient is $\frac{\partial}{\partial \mathbf{w}}(0) = 0$. All together, the total gradient is:

$$\nabla_{\mathbf{w}} = \frac{-1}{N} \sum_{i: (y_i \mathbf{x}_i^T \mathbf{w}) < 1} (y_i \mathbf{x}_i^T) + 2s\mathbf{w}$$

We plot the training loss vs. iteration number in Figure 1.

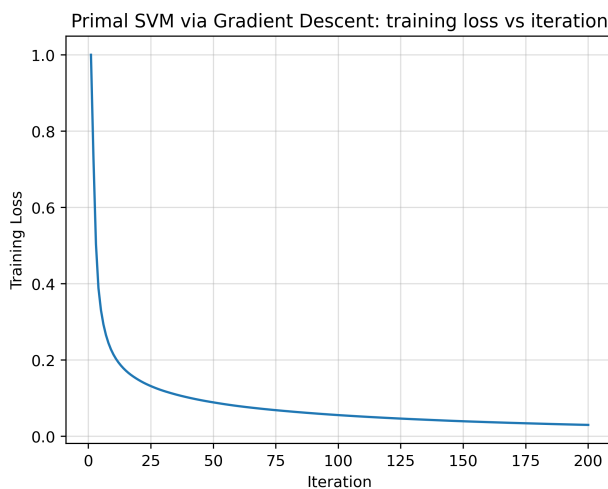


Figure 1: Primal SVM Training Loss vs. Iterations

- (1b) Use SVM package to train linear SVM, $C=1$. We can report the training errors and number of support vectors
- (1c) Use SVM package to train a SVM with polynomial kernel of degree 2.
- (1d) Train an SVM classifier with $C=1$ and RBF kernels γ